

Comparison of DAC, MAC, RBAC, ABAC, and PBAC

Criteria	DAC	MAC	RBAC	ABAC	PBAC
How it works	Owner controls access (ACL-based).	Access based on security labels; users cannot modify permissions.	Access via roles assigned to users.	Access based on attributes (user, resource, environment).	Access controlled by centralized high-level policies.
Domain of application	Personal systems, small networks.	Military, government, high-security systems.	Enterprises, banks, ERP systems.	Cloud, IoT, dynamic systems.	Zero Trust and cloud-native enterprises.
Database changes	Minimal (store ACLs per object).	Requires classification labels for users and objects.	Requires Users, Roles, Permissions, mapping tables.	Requires attribute storage and policy engine.	Requires centralized policy repository and engine.
Centralisation	Low (decentralized).	High (system-enforced).	Centralized role management.	Centralized policy evaluation.	Highly centralized.
Resource usage	Low.	Medium.	Low-Medium.	High (dynamic evaluation).	High.
Complexity	Low.	High.	Medium.	High.	Very High.
Ease of implementation	Easy.	Difficult.	Moderate.	Difficult.	Hard / advanced infrastructure required.
Security level	Low-Medium.	Very High.	High (if well-designed).	Very High.	Extremely High.
Scalability	Poor in large systems.	Medium.	Very Good.	Excellent.	Excellent.
Flexibility	Low.	Low (rigid).	Medium.	Very High.	Extremely High.